Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Measures of Location & Spread

SOLUTIONS

---

1. Mean = 8. **2.** Median = 9. **3.** Mode = 5.
4. Fifth = 13. **5.** Range = 18. **6.** $Q_1 = 7$, $Q_3 = 15$, IQR = 8.
7. Mean = 12. **8.** Variance = 16, SD = 4.
9. Mean = 7. **10.** Mean = $\frac{4\times5+10\times15+6\times25}{20} = 16$.
11. Mean of $y = 0$, SD of $y = 1$. **12.** Mean.
13. 40th percentile ≈ 9 (position 4). **14.** $\sum x = 300$, $\sum x^2 = 4680$.
15. Median more robust; mean pulled by skew/outliers.
16. Mean = 6.4; variance = $\frac{1}{5}(\sum x^2 - 5\bar{x}^2)$; SD ≈ 2.73.
17. Combined mean = $\frac{20\times12+30\times18}{50} = 15.6$.
18. Mean = $3\bar{x} + 2$; SD = $3s$.
19. Mean ≈ 17.2; variance from midpoints.
20. IQR = 13; mean > median suggests slight positive skew.
21. SD = $\sqrt{15} \approx 3.87$. **22.** Mean $x = 62$; variance $x = 64$.
23. Data: 12,15,18,20,21,24,27,33,35; mean/median calculated.
24. e.g. $7, 7, 9, 12, 15$ (check conditions).
25. SD uses all data; less affected by extremes than range.